

TABLE OF SOME LAPLACE TRANSFORMS

$f(t)$	$\mathcal{L}\{f(t)\} = F(s)$	$f(t)$	$\mathcal{L}\{f(t)\} = F(s)$
	$\frac{1}{s}, s > 0$	2. t	$\frac{1}{s^2}$
	$\frac{n!}{s^{n+1}}, s > 0$	4. $t^n, n > -1$	$\frac{\Gamma(n+1)}{s^{n+1}}$
$\frac{1}{\sqrt{t}}$	$\frac{1}{\sqrt{s}}$	6. e^{at}	$\frac{1}{s-a}, s > a$
$\frac{1}{t}$	$\frac{1}{(s-a)^2}$	8. $\sin at$	$\frac{a}{s^2 + a^2}, s > 0$
$\cos at$	$\frac{s}{s^2 + a^2}, s > 0$	10. $\sinh at$	$\frac{a}{s^2 - a^2}, s > a $
$\cosh at$	$\frac{s}{s^2 - a^2}, s > a $	12. $t^n e^{at}$	$\frac{n!}{(s-a)^{n+1}}, s > a$
$e^{at} \sin bt$	$\frac{b}{(s-a)^2 + b^2}, s > a$	14. $e^{at} \cos bt$	$\frac{s-a}{(s-a)^2 + b^2}, s > a$
$e^{at} \cosh bt$	$\frac{2as}{(s^2 + a^2)^2}, s > 0$	16. $t \cos at$	$\frac{s^2 - a^2}{(s^2 + a^2)^2}, s > 0$
$f(ct)$	$\frac{1}{c} F\left(\frac{s}{c}\right), c > 0$	18. $\int_0^t f(u) du$	$\frac{1}{s} F(s)$
$t^n f(t)$	$(-1)^n \frac{d^n}{ds^n} F(s)$	20. $\frac{f(t)}{t}$	$\int_s^\infty F(u) du$
$e^{-at} f(t)$	$\frac{F(s+a)}{s}$	22. $u_c(t) f(t-a)$	$e^{-as} F(s)$
$\sin at - at \cos at$	$\frac{2a^3}{(s^2 + a^2)^2}$	24. $f'(t)$	$s F(s) - f(0)$
$1 - \cos at$	$\frac{a^2}{s(s^2 + a^2)}$	26. $at - \sin at$	$\frac{a^3}{s^2(s^2 + a^2)}$
$\sinh at - \sin at$	$\frac{2a^3}{s^4 - a^4}$	28. $\cosh at - \cos at$	$\frac{2a^3 s}{s^4 - a^4}$

EXERCISE 11.1

Compute the Laplace transform of each of the following (Problems 1–28):

1. $t^2 + 6t - 17$
2. e^{3t}
3. $\sin(7t + 4)$
4. $\cos(at + b)$
5. $\cosh(5t - 3)$
6. $(t^3 - 1)e^{-2t}$
7. $e^{-t} \sin 2t$
8. $e^{2t} \cosh 4t$
9. $\cos t \cos 2t$
10. $\sin^3 t$
11. $te^{-3t} \sin at$
12. $\sinh^2 at$
13. $\cosh at \sin at$
14. $\sinh at \cos at$
15. $\cosh at \cos bt$
16. $[t]$, the bracket function
17. t^α , $\alpha > -1$. Hence find $\mathcal{L}\{t^{\alpha/2}\}$
18. $t^2 \sin at$
19. $t^2 \cos at$
20. $t \sin^2 at$
21. $t^2 \cos^2 2t$
22. $\frac{\sin at}{t}$
23. $\frac{1 - \cos at}{t}$
24. $\int_0^t \frac{\sin au}{u} du$
25. $\int_0^t \frac{1 - \cos au}{u} du$
26. $\frac{\sinh at}{t}$
27. $\ln t$
28.
$$f(t) = \begin{cases} 0 & \text{if } t < 3 \\ (t-3)^3 & \text{if } t > 3 \end{cases}$$
29. If $\mathcal{L}\{f(t)\} = F(s)$ for $s > a$, show that $\mathcal{L}\{f(ct)\} = \frac{1}{c} F\left(\frac{s}{c}\right)$, $c > 0$ and $s > ac$
30. Compute $\mathcal{L}\{\sin \sqrt{t}\}$. Deduce $\mathcal{L}\left\{\frac{\cos \sqrt{t}}{\sqrt{t}}\right\}$

EXERCISE 11.2

compute the inverse Laplace transform of each of the following problems 1 – 20):

$$\frac{-2}{s^2 - 2}$$

$$\frac{9s - 67}{s^2 - 16s + 49}$$

$$\frac{s}{(s+a)^2 + b^2}$$

$$\frac{1}{(s-1)(s^2+4)}$$

$$\frac{5s+3}{(s+7)^5}$$

$$\frac{2s^3 + 6s^2 + 21s + 52}{s(s+2)(s^2+4s+13)}$$

$$\frac{s^3 + 3s^2 - s - 3}{(s^2 + 2s + 5)^2}$$

$$\ln \frac{s^2 + 1}{(s-1)^2}$$

$$\frac{e^{-3s}}{s^2(s^2+9)}$$

$$e^{-2s} \cdot \frac{s+6}{s^3 - 5s^2 + 6s}$$

$$2. \quad \frac{3s+1}{s^2 - 6s + 18}$$

$$4. \quad \frac{as+b}{s^2 + 2cs + d}, \quad d > c^2 > 0$$

$$6. \quad \frac{1}{(s^2 + a^2)(s^2 + b^2)}$$

$$8. \quad \frac{7s+5}{(3s-8)^2}$$

$$10. \quad \frac{2s-3}{2s^3 + 3s^2 - 2s - 3}$$

$$12. \quad \frac{1}{(s^2 + 4)(s^2 + 6s - 5)}$$

$$14. \quad \arctan \frac{a}{s}$$

$$16. \quad \ln \frac{s^2 + a^2}{s^2 + b^2}$$

$$18. \quad e^{-2s} \cdot \frac{s}{s^2 - 4s + 5}$$

$$20. \quad e^{-3s} \cdot \frac{3s-7}{s^2 - 10s + 26}$$

In each of Problems 21 – 23, use the Convolution Theorem 11.20 to evaluate the inverse Laplace transform:

$$\frac{1}{s^2(s+5)}$$

$$22. \quad \frac{s}{(s+1)(s^2+4)}$$